Course Code	Subject	L-T-P	Credits
MN481	**Mineral Processing**	**3-0-2**	**11**

COURSE TOPICS::

Introduction

Importance of minerals and coal in economy of India. Genesis, occurrences of important ore and coal deposits of India. Importance of mineral beneficiation in metallurgical industry.

Liberation and Comminution

Concept and importance of liberation. Theories of Comminution. Crushing and grinding equipment, their fields of application and limitations. Comminution circuits.

Sizing and Classification

Laboratory sizing techniques. Industrial screens – selection and performance. Laws of settling of solids in fluid. Types of classifiers, their selection and performance. Interpretation of sizing data.

Concentration Methods

Principles, equipment and circuits for various concentration processes such as gravity concentration, dense media separation, magnetic separation, high tension separation, flotation. Applications and limitations of each method.

Solid-Liquid Separation

Principles, techniques and application of dewatering units such as thickeners and filters.

Plant Practices

Processing flow sheets for coal and important ores. Metallurgical accounting and control.

READINGS

TEXTBOOKS::

1. B A Wills, Mineral Processing Technology, Elsevier.

2. A M Gaudin, Principle of Mineral Dressing, Tata McGraw-Hill.

3. S K Jain, Mineral Processing, CBS Publisher.